E-Commerce Performance Analysis

Amazon Brazil

SQL Business Case Study

Turning transactional data into business insights and operational recommendations.



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Dataset
Olist Brazilian E-Commerce Dataset

Tools
My SQL (SQL, Analytics, Visualization)

BUSINESS CONTEXT AND DATASET

BUSINESS CONTEXT

Amazon Brazil is scaling rapidly. The operations and strategy team wants to understand customers, sellers, product categories, payments, reviews and delivery performance to make better business decisions.

KEY ANALYSIS AREAS

Revenue & Demand

Which categories grow?
When is demand highest?

Seller Performance

Where are sellers active?
Who are top performers?

Customer Behaviour

How do customers pay and order?
Who are high value customers?

Experience & Operations

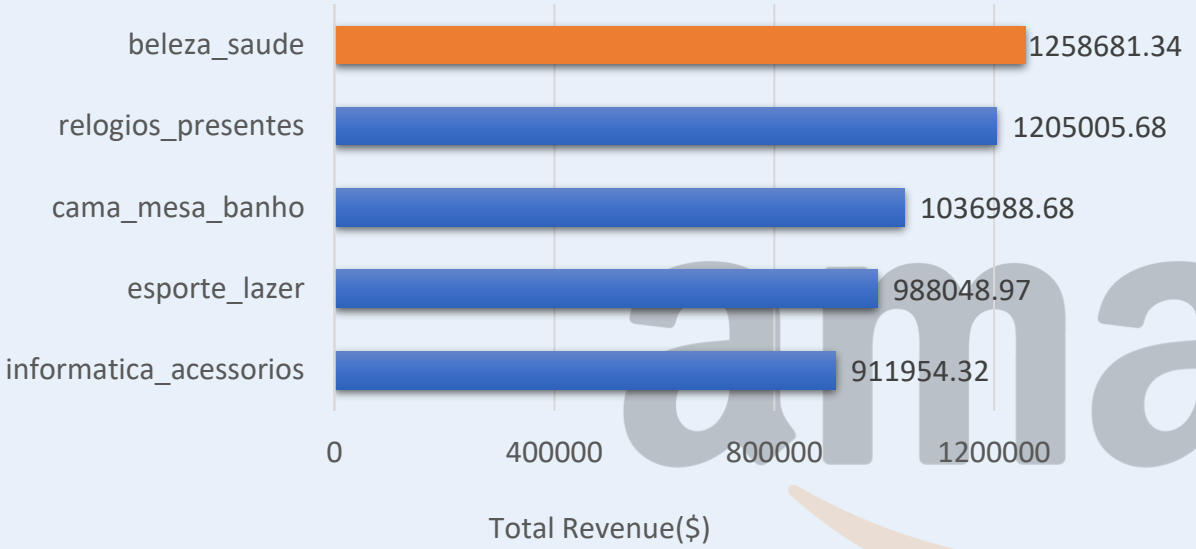
Which categories grow?
When is demand highest?

Dataset Overview (Olist Brazilian E-Commerce Dataset)

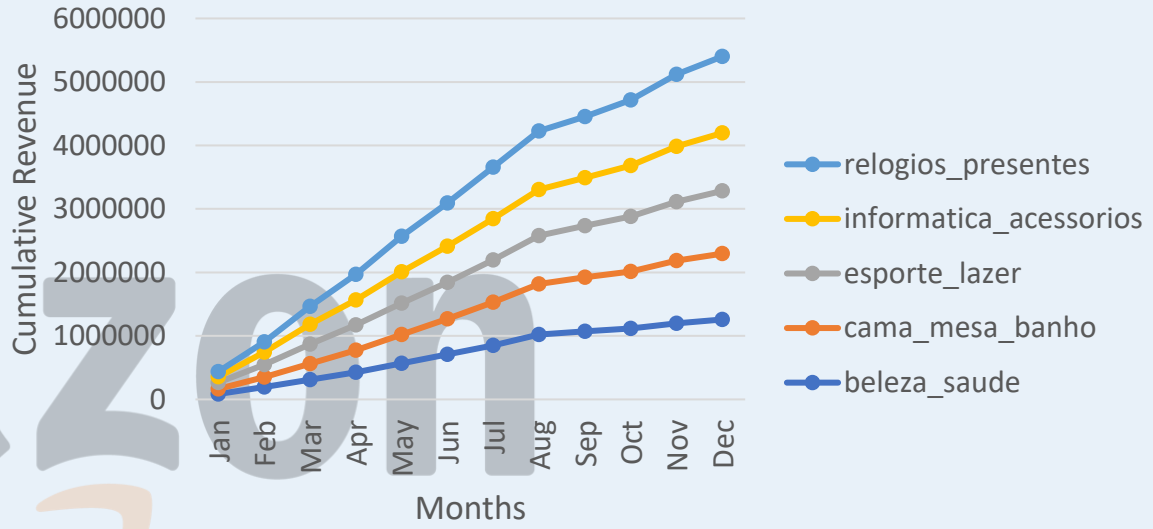
Customers	Orders	Order items	Products	Sellers	Order payments	Order reviews
Customer_id Customer_state	Order_id Order_dates Order_status	Order_id Product_id Seller_id	Product_id Product_category Product_name	seller_id seller_state	order_id Payment_type Payment_value	order_id review_score review_date

REVENUE & CATEGORY PERFORMANCE

Top 5 Categories by Revenue



Cumulative Revenue by Category

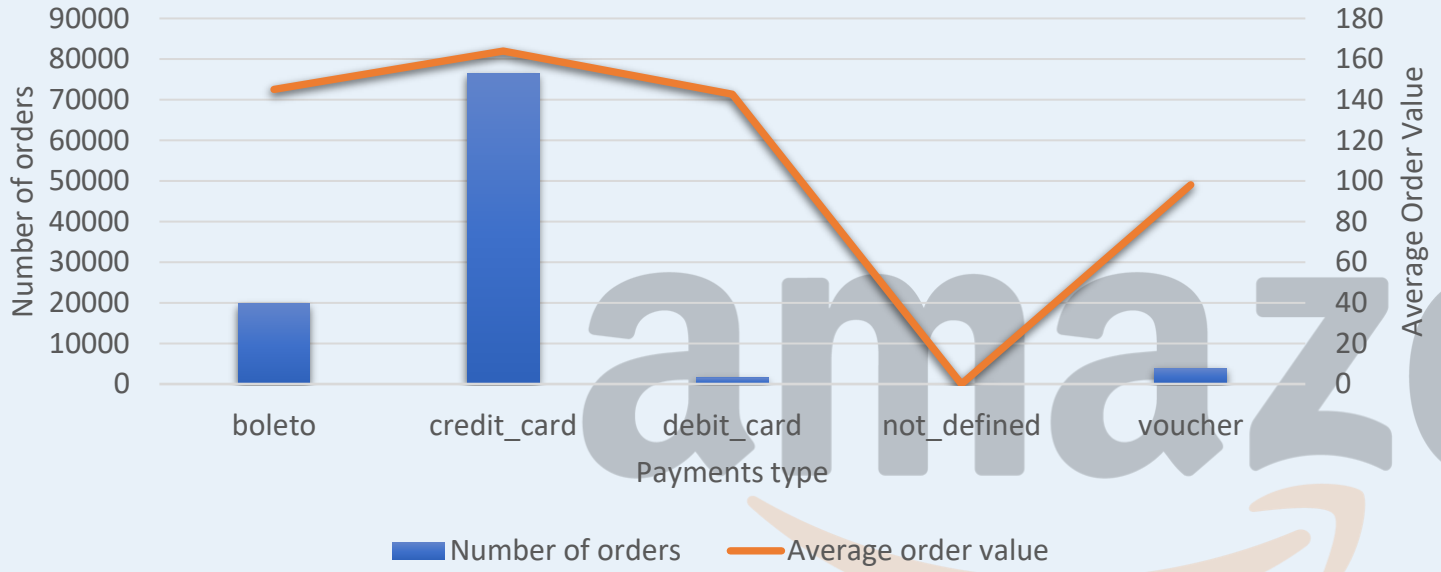



```
SELECT  p.product_category_name,
ROUND(SUM(oi.price), 2)
AS total_revenue
FROM products AS p
JOIN order_items AS oi
            ON p.product_id = oi.product_id
GROUP BY p.product_category_name
ORDER BY total_revenue DESC
LIMIT 5;
```

"In this query, I used a recursive CTE to generate all 12 months. Then I identified the top five product categories based on total revenue. I used JOINS to combine order, product and revenue data, and COALESCE to handle months with no sales. Finally, I applied a window function to calculate the cumulative revenue for each category over the months"

CUSTOMER & PAYMENT METHODS

Most Used Payment Method & Average Order Value



amazo

customer state	total customers	total orders	Avg orders per customer
SP	40302	41746	1.04
RJ	12384	12852	1.04
Others	32191	33208	1.03
MG	11259	11635	1.03

Customer id:
1617b1357756262bf
a56ab541c47bc16

Total Amount Spent

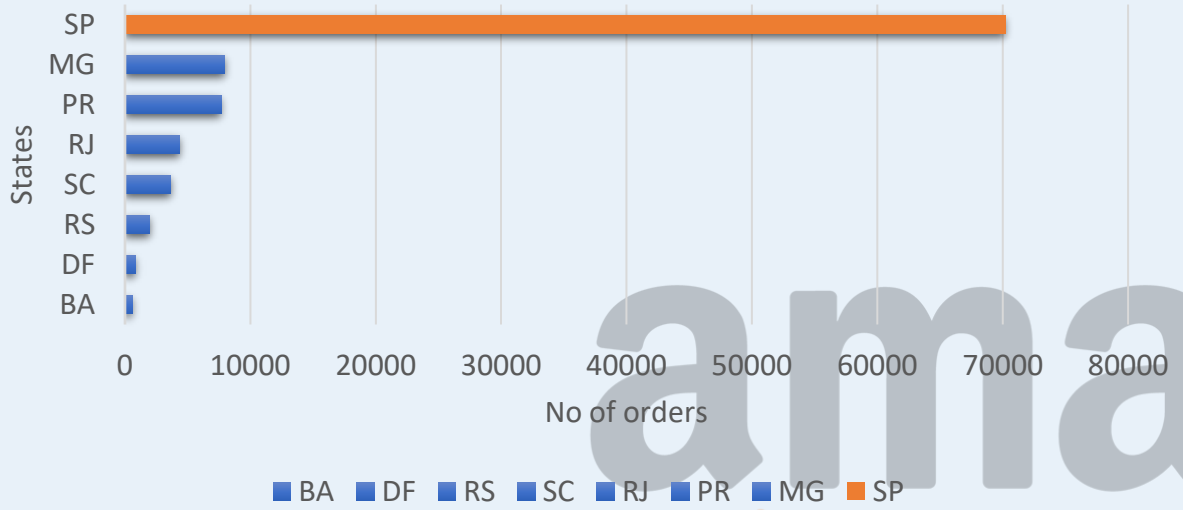
$13,664

Key Insights

- Most used payment: Credit Card
- Highest Average order value: 163.94
- Highest spending Customer:13,664
- State with highest order frequency: SP

Seller Performance & Regional Fulfilment

Orders Fulfilled by Seller State



Sellers with more Orders than Average



Sellers with no Orders

Total Sellers With Zero Orders

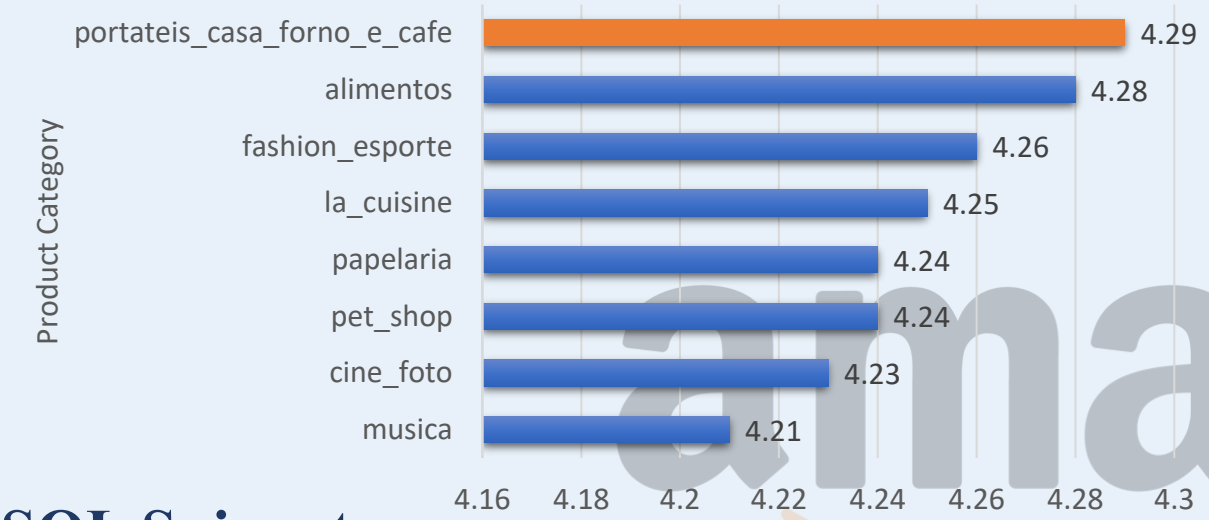
0

Key Insights

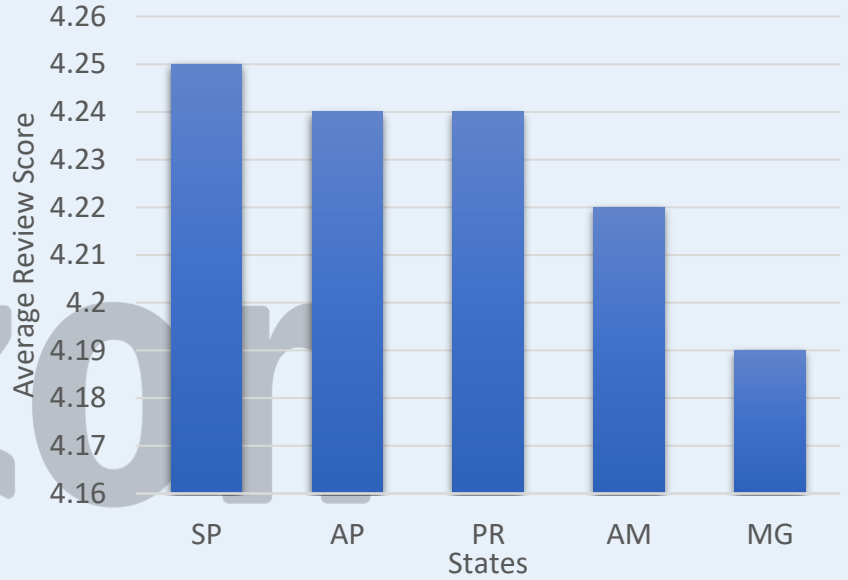
- Top Seller State: SP
- Average order per Seller: 32.31
- No of above Average Sellers:16
- Sellers with no orders: 0

Review Performance & Customer Satisfaction

Avg Review Score by Product Category



Avg Review Score by Destination State



SQL Snippets

```
WITH order_category_reviews AS ( SELECT DISTINCT       order_items.order_id,
products.product_category_name,  order_reviews.review_score    FROM
order_items
JOIN products ON order_items.product_id = products.product_id
JOIN order_reviews ON order_items.order_id = order_reviews.order_id) SELECT
Product_category_name,
ROUND(AVG(review_score), 2) AS Average_review_score
FROM order_category_reviews
GROUP BY product_category_name
ORDER BY average_review_score ASC;
```

Key Insights

- Lowest Scoring Category: seguros e servicos
- Highest Scoring State: SP

Delivery Performance

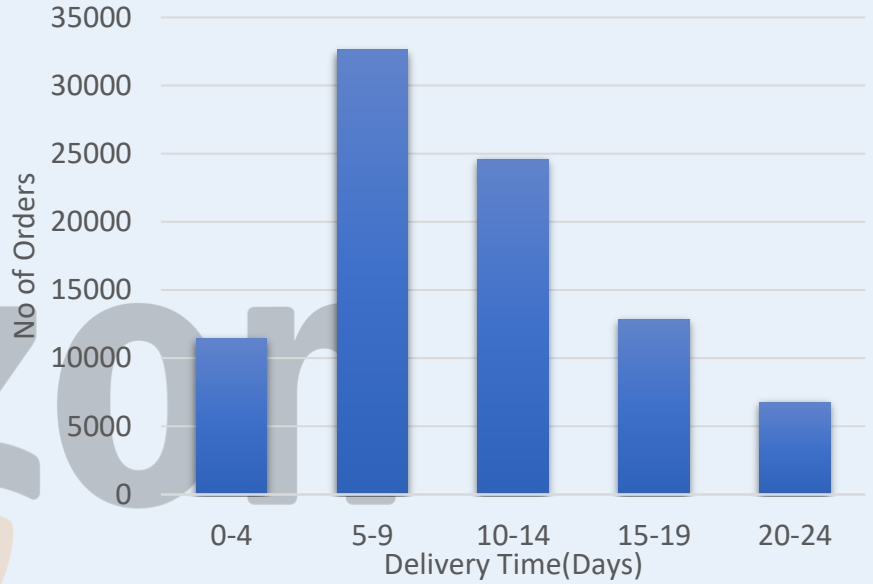
Median Delivery Time

10

Days

Median number of days between order purchase and actual delivery

Distribution of Delivery Time (Days)



SQL Snippets

```
WITH delivery_times AS (    SELECT  order_id, DATEDIFF(
order_delivered_customer_date, order_purchase_timestamp ) AS delivery_days
FROM orders
WHERE order_delivered_customer_date IS NOT NULL  AND
order_purchase_timestamp IS NOT NULL),ranked_delivery AS ( SELECT
delivery_days, ROW_NUMBER() OVER ( ORDER BY delivery_days ) AS rn, COUNT(*)
OVER () AS total_rows
FROM delivery_times)SELECT    AVG(delivery_days) AS median_delivery_days
FROM ranked_delivery WHERE rn IN (FLOOR((total_rows + 1) / 2), CEIL((total_rows +
1) / 2));
```

Key Insights

- Use median to avoid impact of extreme delays
- Compare by state or seller group for deeper analysis.

Monthly Demand Trends

Month with Highest No of Orders



SQL Snippets

```
SELECT    DATE_FORMAT(       order_purchase_timestamp, '%Y-%m' ) AS
order_month,COUNT(*) AS total_orders
FROM orders
WHERE order_purchase_timestamp IS NOT NULL
GROUP BY DATE_FORMAT(order_purchase_timestamp, '%Y-%m')
ORDER BY order_month;
```

Key Insights

- Peak Month: November
- Total Orders: 7544
- Use for Capacity Planning

Catalogue & Customer Opportunities

Products Never Ordered



Total Products

0

Products

Products with no Matching rows in order_items

Customers with No Reviews



Total Customers

30

Customers

Customers who have placed orders but never left a review

Proposed Actions

- Audit listing Quality, Pricing, availability and search visibility before deciding to improve or remove.

Proposed Actions

- Send targeted review reminders
- Test different channels(email, in app)
- Measure review conversion

Key Recommendations



Grow Category Revenue
Use top-category revenue and cumulative trends to select categories for promotional tests.



Improve Fulfillment Planning
Combine seller activity, delivery time and state-level review scores for better logistics decisions.



Enhance Customer Experience
Address low review scores and engage customers who never reviewed.



Strengthen Seller Ecosystem
Support high-performing sellers and re-engage dormant accounts.

THANK YOU